

**Exp no :08**

**Date:**

## **Use Steg-Expose to Detect Hidden Data in Images**

### **Description**

**StegExpose** is a specialized tool used for **steganography analysis**. It works by evaluating the **statistical properties** of an image to estimate the probability of hidden data being embedded within it. This process helps forensic investigators detect steganography techniques like Least Significant Bit (LSB) embedding.

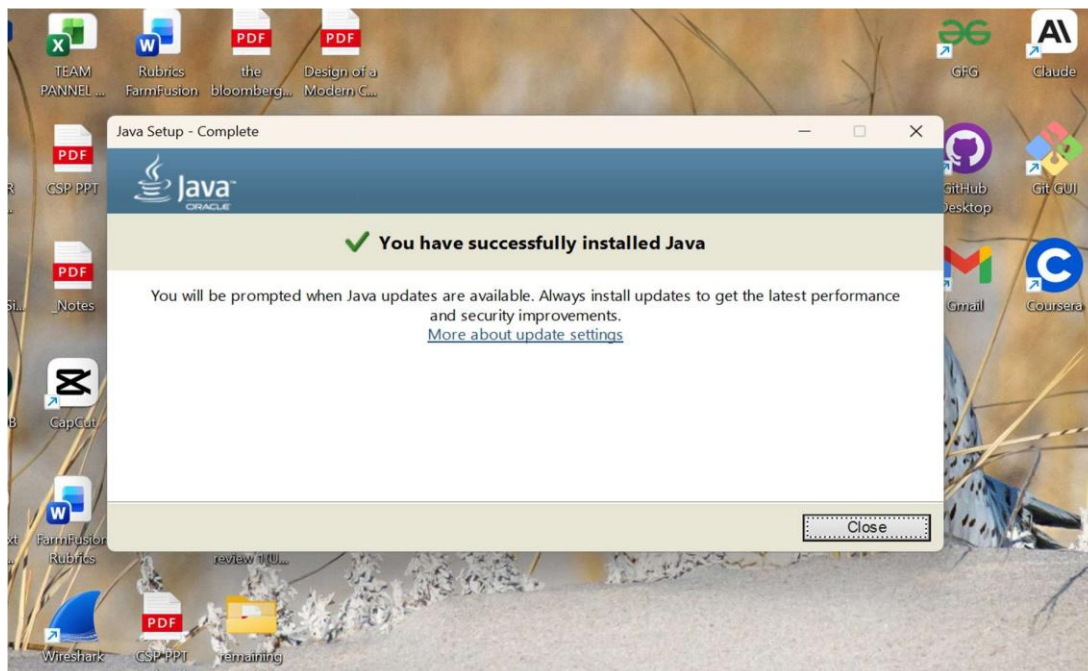
### **Prerequisites**

- Java Runtime Environment (JRE): StegExpose is a Java-based application and requires JRE to run.
- StegExpose Tool: Download the latest .jar file from the official StegExpose GitHub repository.

### **Step-by-Step Process**

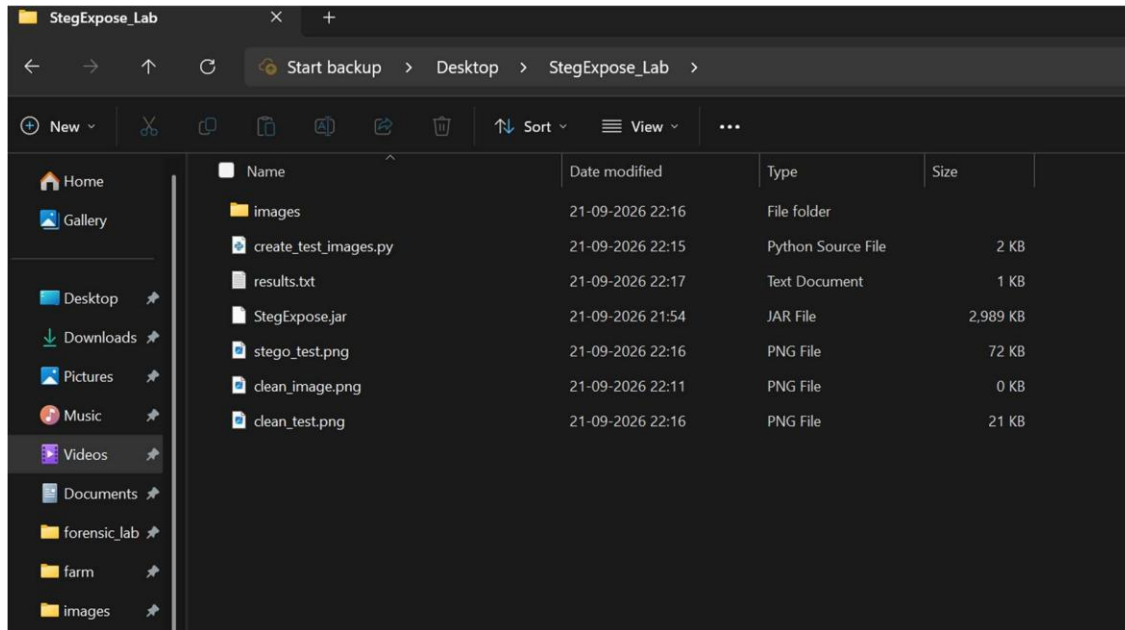
#### **1. Download and Set Up StegExpose**

1. Download the tool: Obtain the StegExpose.jar file from the GitHub repository.
2. Install Java: Ensure JRE is installed on your machine.
3. Prepare environment: Place the StegExpose.jar file in a dedicated working folder.



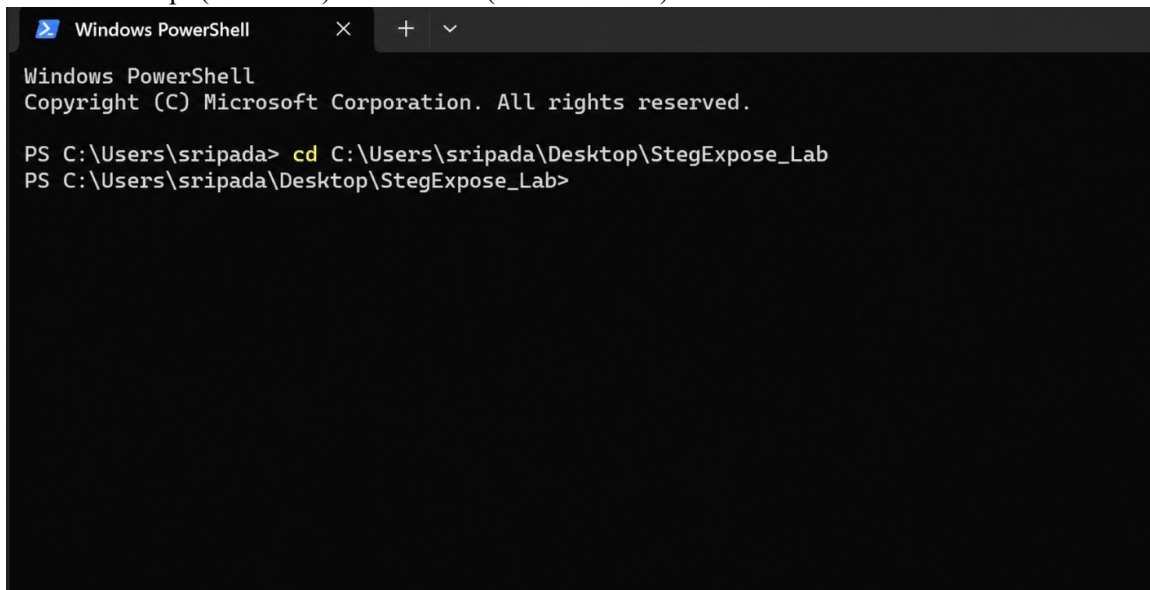
## 2. Select Images for Analysis

- Collect the images you suspect might contain hidden data.
- StegExpose supports common image formats such as .png, .jpg, and .bmp.



## 3. Open Command Line or Terminal

- Navigate to the folder where the StegExpose.jar file is located using your Command Prompt (Windows) or Terminal (Linux/macOS).



#### 4. Run StegExpose on an Image

- Use the following command structure to analyze a single image for hidden data:

##### Command Structure:

```
java -jar StegExpose.jar <image_file_path>
```

##### Example:

```
java -jar StegExpose.jar test_image.png
```

#### 5. Analyze the Output

- StegExpose calculates a **"suspect" score** ranging from 0 to 1. **The higher the score, the more likely steganography is present.**

| Score Range   | Interpretation (Suggested Thresholds)                       |
|---------------|-------------------------------------------------------------|
| Less than 0.2 | Image is considered <b>clean</b> (no hidden data detected). |
| 0.2 - 0.3     | <b>Possibly</b> some hidden data is present.                |
| Above 0.3     | <b>Likely</b> that steganography is present.                |

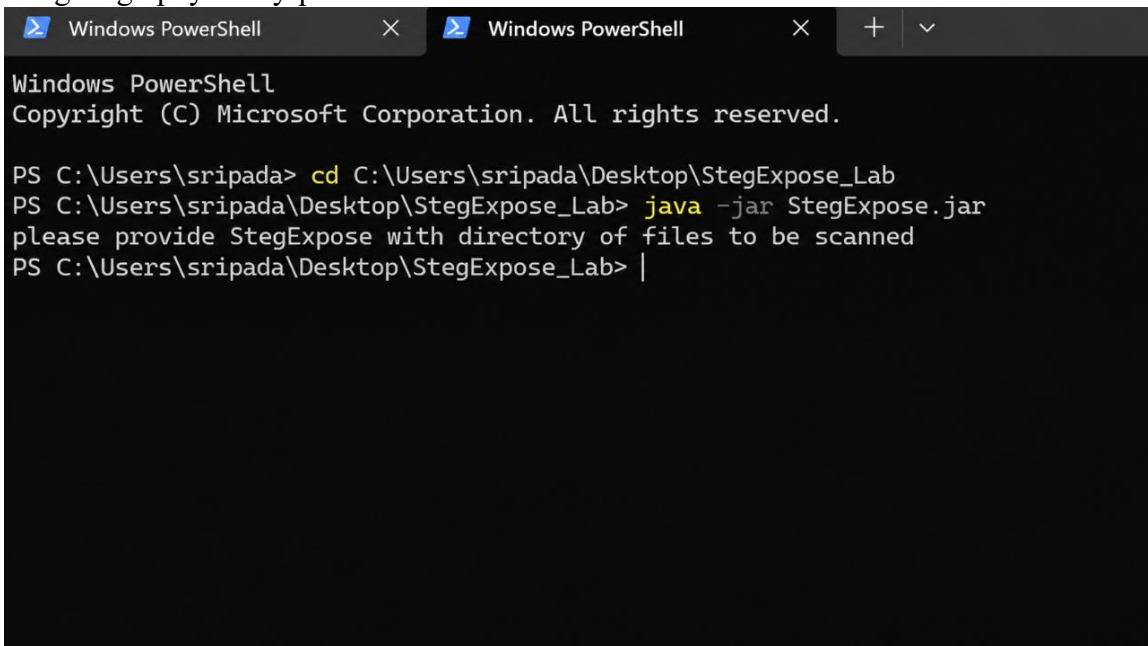
- **Example Output Analysis:**

```
java -jar StegExpose.jar suspect_image.png
```

Analyzing suspect\_image.png...

Result: 0.4

Steganography likely present



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\sripada> cd C:\Users\sripada\Desktop\StegExpose_Lab
PS C:\Users\sripada\Desktop\StegExpose_Lab> java -jar StegExpose.jar
please provide StegExpose with directory of files to be scanned
PS C:\Users\sripada\Desktop\StegExpose_Lab> |
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\sripada> cd C:\Users\sripada\Desktop\StegExpose_Lab
PS C:\Users\sripada\Desktop\StegExpose_Lab> java -jar StegExpose.jar
please provide StegExpose with directory of files to be scanned
PS C:\Users\sripada\Desktop\StegExpose_Lab> java -jar StegExpose.jar images
stego_test.png is suspicious. Approximate amount of hidden data is 5325 bytes.
PS C:\Users\sripada\Desktop\StegExpose_Lab> |
```

## 6. Batch Analysis (Multiple Images)

- To check multiple images at once, specify the folder path containing the images:\

### Command Structure:

```
java -jar StegExpose.jar <folder_path>
```

## 7. Advanced Options (Optional)

- To view additional parameters, such as options for adjusting detection sensitivity or output verbosity, use the --help flag:

### Command:

```
java -jar StegExpose.jar --help
```

## 8. Review the Results

- Review the scores generated for each image and use the threshold values to determine which images require further forensic investigation to extract the suspected hidden data.

### Rubrics

| Criteria & Marks Assigned                              | Mark Allotted | Mark Awarded |
|--------------------------------------------------------|---------------|--------------|
| 1. GitHub Activity & Submission Regularity             | 3             |              |
| 2. Application of Forensic Tools & Practical Execution | 3             |              |
| 3. Documentation & Reporting                           | 2             |              |
| 4. Engagement, Problem-Solving & Team Collaboration    | 2             |              |
| Total                                                  | 10            |              |

### Result:

The hidden data within the image was successfully detected using StegExpose. The analysis revealed that images with a suspect score above the threshold likely contain embedded steganographic content, confirming the tool's effectiveness in steganography detection.